

FTC Robotics Curriculum Outline

Notes (Mostafa):

I used Gemini to generate this, but I did review a lot of it. It still would need some review especially around questions like:

- Does this Task/Resource/Hardware needed combination make sense?
- Better resources? (Especially CAD)
- Is that allotted time actually enough for this task?
- This Task can be replaced, refined, etc.

It references the Alan Smith PDF in the same folder. Especially the Software part.

This curriculum is designed to be a guided, task-based program for First Tech Challenge (FTC) robotics. The goal is to provide students with a series of challenges that encourage independent problem-solving and collaboration, rather than direct instruction.

Software

Resource: [Learn Java for FTC \(PDF\)](#), along with other provided links.

Software.Course1: Introduction to Java for FTC (10 Sessions)

Goal: To establish a strong foundation in Java programming concepts and the basic structure of an FTC robot program.

- **Software.Course1.Session1 (Est. 2 hours):** Introduction to the Development Environment and Hardware
 - **Task 1:** Getting Started
 - **Goal:** Successfully set up the development environment.
 - **Time Estimate:** 30 minutes
 - **Hardware Needed:** Laptop or desktop computer.
 - **Resource:** [Learn Java for FTC](#) PDF, "Introduction" (p. 1) and "Now you try" (p. 6).
 - **Task 2:** Understand the Driver Station
 - **Goal:** Learn how the Driver Station (Android phone) interacts with the Control Hub and displays telemetry.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Driver Station phone with the app installed.
 - **Resource:** [Learn Java for FTC](#) PDF, "Driver Station" (section 1.1.3, p. 2) and "Hello, World" (p. 3). (Resource: [FTC-docs](#))
 - **Task 3:** Variables and Data
 - **Goal:** Create variables for motor power and sensor readings, and display their values on the Driver Station.
 - **Time Estimate:** 30 minutes

- **Hardware Needed:** Driver Station phone, Control Hub.
 - **Resource:** [Learn Java for FTC](#) PDF, "Variables and Data Types" (Ch. 2, p. 13).
- **Software.Course1.Session2 (Est. 2.5 hours):** Basic Programming Concepts
 - **Task 1:** Perform calculations
 - **Goal:** Write code to scale a sensor reading to a desired motor power value.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Driver Station phone, Control Hub.
 - **Resource:** [Learn Java for FTC](#) PDF, "Basic Math" (section 3.1, p. 18).
 - **Task 2:** Basic Control Flow
 - **Goal:** Use an `if/else` statement to change the robot's behavior based on a condition, such as a gamepad button press.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Driver Station phone, Gamepad.
 - **Resource:** [Learn Java for FTC](#) PDF, "Making decisions" (Ch. 4, p. 21).
- **Software.Course1.Session3 (Est. 2.5 hours):** Gamepad Control
 - **Task 1:** Create a loop
 - **Goal:** Use a `while` loop to continuously read and update sensor values or motor power while the OpMode is running.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Driver Station phone, Control Hub.
 - **Resource:** [Learn Java for FTC](#) PDF, "While" (section 4.4, p. 25).
 - **Task 2:** Gamepad and Tele-Op
 - **Goal:** Write a program that reads input from the gamepad's joysticks and buttons and displays the values on the telemetry.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Driver Station phone, Gamepad.
 - **Resource:** [Learn Java for FTC](#) PDF, "Gamepad and basic math" (Ch. 3, p. 17).
- **Software.Course1.Session4 (Est. 1.5 hours):** Controlling a Single Motor
 - **Task 1:** Control a single motor
 - **Goal:** Use gamepad input to control the speed and direction of one motor.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Driver Station phone, Gamepad, Control Hub, one motor.
 - **Resource:** [Learn Java for FTC](#) PDF, "Motors" (Ch. 7, p. 49).
- **Software.Course1.Session5 (Est. 1.5 hours):** Motors and Hardware Mapping
 - **Task 1:** Motors and Hardware Mapping
 - **Goal:** Create a `HardwareMap` to correctly identify and access your robot's motors, servos, and sensors.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Driver Station phone, Control Hub.
 - **Resource:** [Learn Java for FTC](#) PDF, "Configuration file" (section 6.1, p. 41) and "HardwareMap" (p. 44).
- **Software.Course1.Session6 (Est. 3 hours):** Drivetrain and `LinearOpMode`
 - **Task 1:** Control a four-motor drivetrain
 - **Goal:** Use gamepad input to control the speed and direction of a standard four-motor tank or arcade drive.
 - **Time Estimate:** 2 hours
 - **Hardware Needed:** Driver Station phone, Gamepad, Control Hub, four motors.

- **Resource:** [Learn Java for FTC](#) PDF, "4 motor mecanum drive" (section 20.2, p. 142).
 - **Task 2:** Driving Basics and LinearOpMode
 - **Goal:** Write a program that uses motor encoders to drive the robot a specific distance.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Driver Station phone, Gamepad, Control Hub, four motors with encoders.
 - **Resource:** [Learn Java for FTC](#) PDF, "Motor as Sensor" (section 7.4, p. 53).
- **Software.Course1.Session7 (Est. 2.5 hours):** Subsystems and Servos
 - **Task 1:** Understand [LinearOpMode](#)
 - **Goal:** Convert a basic OpMode into a [LinearOpMode](#) and understand the differences in structure and flow.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Driver Station phone, Control Hub.
 - **Resource:** [Learn Java for FTC](#) PDF, "Linear OpMode" (Appendix B, p. 183).
 - **Task 2:** Introduction to Subsystems
 - **Goal:** Write a class that abstracts the control of a simple mechanism (e.g., a single servo claw) and its associated motors or servos.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Driver Station phone, Control Hub.
 - **Resource:** [Learn Java for FTC](#) PDF, "Classes and Objects" (p. 89) and "Creating your own classes" (section 5.4, p. 34).
 - **Task 3:** Servos and Sensors
 - **Goal:** Write an OpMode that uses a gamepad button to move a servo between two predefined positions.
 - **Time Estimate:** 30 minutes
 - **Hardware Needed:** Driver Station phone, Gamepad, Control Hub, one servo.
 - **Resource:** [Learn Java for FTC](#) PDF, "Servos" (Ch. 8, p. 59).
- **Software.Course1.Session8 (Est. 2.5 hours):** Integrating and Testing
 - **Task 1:** Integrate your subsystem
 - **Goal:** Add the subsystem you created into your main tele-op program and control it with the gamepad.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Driver Station phone, Gamepad, Control Hub, at least one motor and servo.
 - **Resource:** [Learn Java for FTC](#) PDF, "Mechanisms" (section 6.2, p. 43).
 - **Task 2:** Read a sensor
 - **Goal:** Use a color, distance, or touch sensor to make a decision in your code (e.g., stop the robot when it is a certain distance from a wall).
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Driver Station phone, Control Hub, a sensor (color, distance, or touch).
 - **Resource:** [Learn Java for FTC](#) PDF, "Analog Sensors" (Ch. 9, p. 63) and "Color and Distance Sensors" (Ch. 10, p. 67).
- **Software.Course1.Session9 (Est. 2.5 hours):** Autonomous Fundamentals
 - **Task 1:** Autonomous Fundamentals
 - **Goal:** Write a simple autonomous program that uses [getRuntime\(\)](#) to move the robot forward for a set amount of time.

- **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Driver Station phone, Control Hub, motors.
 - **Resource:** [Learn Java for FTC](#) PDF, "It's all relative" (section 12.3, p. 86).
- **Task 2:** Use a basic state machine
 - **Goal:** Create a simple state machine to control a sequence of autonomous actions (e.g., drive, stop, turn).
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Driver Station phone, Control Hub, motors.
 - **Resource:** [Learn Java for FTC](#) PDF, "Dealing with State" (Ch. 12, p. 77) and "Autonomous state" (section 12.2, p. 79).
- **Software.Course1.Session10 (Est. 2 hours):** Code Review and Debugging
 - **Task 1:** Code Review and Debugging
 - **Goal:** Read through your code and refactor it for clarity, adding comments and organizing methods.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer.
 - **Resource:** [Learn Java for FTC](#) PDF, "Comments" (section 1.4, p. 8) and "Javadoc" (Ch. 17, p. 127).
 - **Task 2:** Debug a bug
 - **Goal:** Intentionally introduce an error into your code and use the debugger to find and fix it.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer, Driver Station phone, Control Hub.
 - **Resource:** [Learn Java for FTC](#) PDF, "Gotchas" (section 1.6, p. 10).

Software.Course2: Advanced Robotics Programming (10 Sessions)

Goal: To teach advanced programming techniques, including motion control, state machines, and external libraries.

- **Software.Course2.Session1 (Est. 3 hours):** Mecanum Drive
 - **Task 1:** Introduction to Mecanum Drive
 - **Goal:** Explain how Mecanum wheels work and how to calculate motor powers for field-centric or robot-centric movement.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer, whiteboard/paper for demonstration.
 - **Resource:** [Learn Java for FTC](#) PDF, "4 motor mecanum drive" (section 20.2, p. 142) and "Holonomic Drive" (p. 142). (Resource: GM0.org's [Mecanum Drive Tutorial](#) (<https://gm0.org/en/latest/docs/software/tutorials/mecanum-drive.html>))
 - **Task 2:** Program a Mecanum drivetrain
 - **Goal:** Write a tele-op program for a Mecanum robot that allows for strafing and rotation.
 - **Time Estimate:** 2 hours
 - **Hardware Needed:** Driver Station phone, Gamepad, Control Hub, four mecanum wheels with motors.
 - **Resource:** [Learn Java for FTC](#) PDF, "Mecanum Mechanism" (section 20.2.1, p. 143).
- **Software.Course2.Session2 (Est. 3 hours):** Road Runner Basics
 - **Task 1:** Road Runner Basics

- **Goal:** Integrate the Road Runner library into your FTC project and understand its basic structure for autonomous paths.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a stable internet connection.
 - **Resource:** The [Road Runner Docs](https://rr.brott.dev/docs/) (<https://rr.brott.dev/docs/>).
 - **Task 2:** Drive a simple path
 - **Goal:** Use a [TrajectorySequence](#) to program the robot to drive in a straight line or a simple curve.
 - **Time Estimate:** 2 hours
 - **Hardware Needed:** Driver Station phone, Control Hub, motors with encoders.
 - **Resource:** The [Road Runner Docs](https://rr.brott.dev/docs/) (<https://rr.brott.dev/docs/>).
- **Software.Course2.Session3 (Est. 3 hours):** MeepMeep and Advanced Trajectories
 - **Task 1:** MeepMeep Visualization
 - **Goal:** Use the MeepMeep tool to visually create a complex autonomous path that includes turns and curves.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** The [MeepMeep](#) section on the Road Runner website (<https://learnroadrunner.com/tool/meepmeep.html>).
 - **Task 2:** Transfer the trajectory
 - **Goal:** Convert the code from MeepMeep into your robot's Road Runner program.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer.
 - **Resource:** [Road Runner Docs](#) (<https://rr.brott.dev/docs/>).
- **Software.Course2.Session4 (Est. 3 hours):** Pathing and State Machines
 - **Task 1:** Advanced Trajectories
 - **Goal:** Create an autonomous path that uses splines for smooth curves and precise turns to avoid obstacles.
 - **Time Estimate:** 2 hours
 - **Hardware Needed:** Driver Station phone, Control Hub, motors with encoders.
 - **Resource:** The [Road Runner Docs](https://rr.brott.dev/docs/) (<https://rr.brott.dev/docs/>).
 - **Task 2:** Follow a sequence of actions
 - **Goal:** Program the robot to follow a sequence of different trajectories to reach multiple waypoints on the field.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Driver Station phone, Control Hub, motors with encoders.
 - **Resource:** The [Road Runner Docs](https://rr.brott.dev/docs/) (<https://rr.brott.dev/docs/>).
- **Software.Course2.Session5 (Est. 2.5 hours):** Advanced State Machines and Subsystems
 - **Task 1:** Advanced State Machines
 - **Goal:** Create a more robust state machine for a multi-step autonomous program that includes sensor feedback to transition between states.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Driver Station phone, Control Hub, motors, sensors.
 - **Resource:** [Learn Java for FTC](#) PDF, "Dealing with State" (Ch. 12, p. 77) and "Using the switch statement" (section 12.2.1, p. 81).
 - **Task 2:** Integrate subsystems with the state machine
 - **Goal:** Control a lift or intake subsystem from within your autonomous state machine.
 - **Time Estimate:** 1 hour

- **Hardware Needed:** Driver Station phone, Control Hub, motors, and servos for the subsystem.
 - **Resource:** [Learn Java for FTC](#) PDF, "Mechanisms" (section 6.2, p. 43).
- **Software.Course2.Session6 (Est. 3 hours):** Advanced Sensors and Control
 - **Task 1:** Advanced Sensors
 - **Goal:** Use the robot's IMU to get its heading and use that data to achieve accurate turns.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Driver Station phone, Control Hub with IMU.
 - **Resource:** [Learn Java for FTC](#) PDF, "Gyro (IMU)" (Ch. 11, p. 73) and "IMU" (p. 75).
 - **Task 2:** Implement color detection
 - **Goal:** Use a color sensor to detect a game element and make a decision based on its color.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Driver Station phone, Control Hub, a color sensor.
 - **Resource:** [Learn Java for FTC](#) PDF, "Color and Distance Sensors" (Ch. 10, p. 67).
- **Software.Course2.Session7 (Est. 3 hours):** Control Theory
 - **Task 1:** Control Theory Fundamentals
 - **Goal:** Explain the basics of PID control and tune the **P** and **D** gains for a motor to get it to hold a position accurately.
 - **Time Estimate:** 2 hours
 - **Hardware Needed:** Driver Station phone, Control Hub, motor with an encoder.
 - **Resource:** [Learn Java for FTC](#) PDF, "PID Control" (Ch. 24, p. 165).
 - **Task 2:** Implement a simple feedback loop
 - **Goal:** Use a sensor (e.g., an encoder) to create a feedback loop that corrects for motor slippage.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Driver Station phone, Control Hub, motor with an encoder.
 - **Resource:** [Learn Java for FTC](#) PDF, "Motor as Sensor" (section 7.4, p. 53).
- **Software.Course2.Session8 (Est. 3 hours):** Vision Processing
 - **Task 1:** Vision Processing
 - **Goal:** Explain the basics of computer vision and how it can be used in FTC.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer, webcam or phone with camera.
 - **Resource:** [Learn Java for FTC](#) PDF, "Computer Vision" (Ch. 16, p. 109) and "OpenCV" (p. 109).
 - **Task 2:** Use a vision library
 - **Goal:** Implement a simple vision pipeline using a library like EasyOpenCV to detect the location of an object on the field.
 - **Time Estimate:** 2 hours
 - **Hardware Needed:** Driver Station phone with camera, Control Hub.
 - **Resource:** [Learn Java for FTC](#) PDF, "April Tags" (section 16.1, p. 109) and "Actual computer vision" (section 16.5, p. 122).
- **Software.Course2.Session9 (Est. 2.5 hours):** Object-Oriented Design
 - **Task 1:** Code Refactoring and Object-Oriented Design
 - **Goal:** Refactor your **TeleOp** and **Autonomous** programs to use separate classes for each subsystem (drivetrain, lift, intake).
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer.

- **Resource:** [Learn Java for FTC](#) PDF, "Class Members and Methods" (Ch. 5, p. 29) and "Creating your own classes" (section 5.4, p. 34).
 - **Task 2:** Use inheritance
 - **Goal:** Create a base [OpMode](#) class to share common methods and variables across your tele-op and autonomous programs.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer.
 - **Resource:** [Learn Java for FTC](#) PDF, "Inheritance" (Ch. 14, p. 95).
 - **Software.Course2.Session10 (Est. 2.5 hours):** Git and Version Control
 - **Task 1:** Git and Version Control
 - **Goal:** Initialize a Git repository for your team's code and make your first commit.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer.
 - **Resource:** [A simple guide to Git for beginners](#).
 - **Task 2:** Collaborate with Git
 - **Goal:** Push your code to a shared repository and pull in changes from another teammate.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer.
 - **Resource:** [Git & GitHub Crash Course For Beginners](#).
-

CAD

Resource: The provided Onshape links.

CAD.Course1: Introduction to Onshape (10 Sessions)

Goal: To become proficient with the Onshape interface and the fundamental principles of CAD for robotics.

- **CAD.Course1.Session1 (Est. 1 hour):** Onshape Basics and Interface
 - **Task 1:** Onshape Basics
 - **Goal:** Successfully set up a free Onshape account and navigate the dashboard.
 - **Time Estimate:** 30 minutes
 - **Hardware Needed:** Computer with a web browser.
 - **Task 2:** Learn the interface
 - **Goal:** Create a new document, learn to navigate the 3D space, and understand the basic tools like Sketch, Extrude, and Revolve.
 - **Time Estimate:** 30 minutes
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape Learning Path: [Introduction to Onshape](#) (<https://learn.onshape.com/courses/intro-to-onshape>) and [Intro to Onshape](#) (<https://learn.onshape.com/learn/article/intro-to-onshape>)
- **CAD.Course1.Session2 (Est. 2 hours):** Basic Part Modeling
 - **Task 1:** Basic Part Modeling
 - **Goal:** Use sketch tools to create a simple shape like a square or a circle with specific dimensions.
 - **Time Estimate:** 1 hour

- **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape Learning Path: [Introduction to CAD](https://learn.onshape.com/learning-paths/introduction-to-cad) (<https://learn.onshape.com/learning-paths/introduction-to-cad>)
 - **Task 2:** Turn your sketch into a 3D part
 - **Goal:** Use the **Extrude** feature to give depth to your 2D sketch, creating a simple 3D part.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape Learning Path: [Introduction to CAD](https://learn.onshape.com/learning-paths/introduction-to-cad) (<https://learn.onshape.com/learning-paths/introduction-to-cad>)
- **CAD.Course1.Session3 (Est. 2 hours):** Complex Parts and Features
 - **Task 1:** More Complex Parts
 - **Goal:** Create a part that requires multiple sketches and extrusions on different planes.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** [Learning the basics of working with multiple sketches in Onshape CAD](#)
 - **Task 2:** Use the **Hole** and **Fillet** features
 - **Goal:** Add holes and rounded edges to your part for a more finished look.
 - **Time Estimate:** 30 minutes
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** [Onshape Hole Feature Help](#) and [Onshape Fillet Feature Help](#).
- **CAD.Course1.Session4 (Est. 2.5 hours):** Introduction to Assemblies
 - **Task 1:** First Assembly
 - **Goal:** Use the **Insert** feature to add standard FTC components from the Onshape FTC library into a new assembly.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape **FTC Catalog** (<https://learn.onshape.com/catalog?query=ftc>)
 - **Task 2:** Use **Mate** connectors
 - **Goal:** Use **Mate** features like "Fasten" and "Revolute" to connect two parts together.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape Article: [Assemble a Robot](https://learn.onshape.com/learn/article/assemble-a-robot) (<https://learn.onshape.com/learn/article/assemble-a-robot>)
- **CAD.Course1.Session5 (Est. 2.5 hours):** Mates and Simulation
 - **Task 1:** Understanding Mates
 - **Goal:** Experiment with different mate types (**Revolute**, **Slider**, **Cylindrical**) to understand how they restrict movement.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape Article: [Assemble a Robot](https://learn.onshape.com/learn/article/assemble-a-robot) (<https://learn.onshape.com/learn/article/assemble-a-robot>)
 - **Task 2:** Simulate a simple mechanism
 - **Goal:** Create an assembly of two parts that can move relative to each other using a **Mate** connector.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** [Onshape Assembly Simulation Help](#).

- **CAD.Course1.Session6 (Est. 3 hours):** Drivetrain CAD - Chassis
 - **Task 1:** Drivetrain CAD - Chassis
 - **Goal:** Create a new Part Studio and design the main frame of a four-wheeled drivetrain.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** [Onshape for Robotics - Design of a Robot Chassis](#) and [Onshape Robot Tutorial for Absolute Beginners](#).
 - **Task 2:** Assemble the chassis
 - **Goal:** Assemble the chassis components you designed and make sure they are properly mated.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** [Onshape Strafer Chassis Tutorial](#) and [CAD for Robotics - Assemble a Drivetrain](#).
- **CAD.Course1.Session7 (Est. 3 hours):** Drivetrain CAD - Motors and Wheels
 - **Task 1:** Drivetrain CAD - Motors and Wheels
 - **Goal:** Add motors from the FTC Onshape library and mate them correctly to your chassis.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** [Onshape Strafer Chassis Tutorial](#).
 - **Task 2:** Add wheels
 - **Goal:** Add standard FTC wheels and mate them to the motor output shafts.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** [FTC Drivetrain CAD feedback - Chief Delphi](#).
- **CAD.Course1.Session8 (Est. 2 hours):** Subsystem Design
 - **Task 1:** Subsystem Concepts
 - **Goal:** Choose a simple FTC subproblem (e.g., a simple lift or a claw) and sketch out a design concept.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** [FTC Game Manual 0: Design Strategy](#)
 - **Task 2:** Design a custom bracket
 - **Goal:** Create a simple custom bracket in Onshape to connect two parts of your planned subsystem.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape Article: [Creating Custom Components](#) (<https://learn.onshape.com/learn/article/cad-for-robotics-creating-custom-components>)
- **CAD.Course1.Session9 (Est. 2 hours):** Collaboration
 - **Task 1:** Sharing and Collaboration
 - **Goal:** Share an Onshape document with a teammate and grant them editing permissions.
 - **Time Estimate:** 30 minutes
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape Article: [CAD for Robotics Collaboration](#) (<https://learn.onshape.com/learn/article/cad-for-robotics-collaboration>)
 - **Task 2:** Comment on a design

- **Goal:** Use the comment feature in Onshape to provide feedback on a teammate's design.
 - **Time Estimate:** 30 minutes
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape Article: [CAD for Robotics Collaboration](https://learn.onshape.com/learn/article/cad-for-robotics-collaboration) (<https://learn.onshape.com/learn/article/cad-for-robotics-collaboration>)
- **Task 3:** Manufacturing and Documentation
 - **Goal:** Create a basic drawing and Bill of Materials (BOM) for a custom part.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape Article: [Communicating Your Design](https://learn.onshape.com/learn/article/cad-for-robotics-communicating-your-design) (<https://learn.onshape.com/learn/article/cad-for-robotics-communicating-your-design>)
- **CAD.Course1.Session10 (Est. 2 hours):** Prepare for Manufacturing
 - **Task 1:** Prepare for manufacturing
 - **Goal:** Export a part file in a format suitable for manufacturing (e.g., DXF for laser cutting, STL for 3D printing).
 - **Time Estimate:** 2 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape Article: [Manufacturing](https://learn.onshape.com/learn/article/cad-for-robotics-manufacturing) (<https://learn.onshape.com/learn/article/cad-for-robotics-manufacturing>)

CAD.Course2: Advanced CAD (10 Sessions)

Goal: To learn advanced modeling techniques, design for manufacturing, and professional collaboration workflows.

- **CAD.Course2.Session1 (Est. 3 hours):** Design for Manufacturing
 - **Task 1:** Design for Manufacturing (DFM)
 - **Goal:** Explain the basic principles of DFM, focusing on how design choices affect manufacturing processes like 3D printing and laser cutting.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer.
 - **Resource:** [FTC Game Manual 0: Design Strategy](#)
 - **Task 2:** Create a laser-cut part
 - **Goal:** Design a custom part (e.g., a chassis plate) that can be easily laser-cut from a flat sheet of material.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Task 3:** Create a 3D-printed part
 - **Goal:** Design a custom part that uses features like overhangs and support structures appropriate for 3D printing.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
- **CAD.Course2.Session2 (Est. 2.5 hours):** Advanced Features
 - **Task 1:** Use Advanced Features - [Sweep](#)
 - **Goal:** Use the [Sweep](#) feature to create a part with a profile that follows a complex curve.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Task 2:** Use Advanced Features - [Loft](#)

- **Goal:** Use the **Loft** feature to create a part by blending two or more profiles together.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
- **CAD.Course2.Session3 (Est. 2.5 hours):** Pattern and Move
 - **Task 1:** Use Advanced Features - **Pattern**
 - **Goal:** Use the **Pattern** feature (linear or circular) to quickly create multiple instances of a feature or part.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Task 2:** Use Advanced Features - **Move Face**
 - **Goal:** Use the **Move Face** tool to make quick edits to an existing part without having to change the original sketches.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
- **CAD.Course2.Session4 (Est. 2 hours):** Assembly Simulation
 - **Task 1:** Assembly Simulation - Motion
 - **Goal:** Create a simple gear train assembly and use **Mate** connectors to simulate its motion.
 - **Time Estimate:** 2 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** [Onshape Assembly Simulation Help](#).
- **CAD.Course2.Session5 (Est. 3 hours):** Featurescripts and Versioning
 - **Task 1:** Use Featurescripts
 - **Goal:** Find and use a Featurescript from the Onshape community to automate a complex task, such as generating gears or sprockets.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape Learning Center: [FeatureScript Fundamentals](#).
 - **Task 2:** Create a Version
 - **Goal:** Create a named version of your CAD model at a major milestone (e.g., "Drivetrain Complete").
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape **Versions and History** (<https://learn.onshape.com/learn/article/introduction-to-onshape-versions-and-history>).
- **CAD.Course2.Session6 (Est. 2.5 hours):** Documentation
 - **Task 1:** Bill of Materials (BOM)
 - **Goal:** Generate a detailed Bill of Materials for your final robot assembly.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape **Communicating Your Design** (<https://learn.onshape.com/learn/article/cad-for-robotics-communicating-your-design>).
 - **Task 2:** Create a Technical Drawing
 - **Goal:** Create a 2D technical drawing of a custom part with dimensions, tolerances, and a title block.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.

- **Resource:** Onshape [Communicating Your Design](https://learn.onshape.com/learn/article/cad-for-robotics-communicating-your-design) (<https://learn.onshape.com/learn/article/cad-for-robotics-communicating-your-design>).
- **CAD.Course2.Session7 (Est. 2.5 hours):** Configurations and Subsystems
 - **Task 1:** Use the Onshape [Configurations](#) tool
 - **Goal:** Create a configurable part or assembly that can be easily modified (e.g., a chassis with different lengths).
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape [Configurations Help](#) (<https://cad.onshape.com/help/Content/configurations.htm>).
 - **Task 2:** Design an Intake Mechanism
 - **Goal:** Design and model a subsystem, such as an intake or a lift, from scratch in a new Part Studio.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** [Designing Our FTC Bot 2023-2024 - Instructables](#)
- **CAD.Course2.Session8 (Est. 2.5 hours):** Assembly Visualization
 - **Task 1:** Assemble the Intake
 - **Goal:** Assemble the intake subsystem with the drivetrain, ensuring all mates work correctly.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Task 2:** Create Exploded View
 - **Goal:** Create an exploded view of a complex assembly to show how the parts fit together.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape [Exploded Views Help](#) (<https://cad.onshape.com/help/Content/exploded-views.htm>).
- **CAD.Course2.Session9 (Est. 2.5 hours):** Rendering and Collaboration
 - **Task 1:** Render an Assembly
 - **Goal:** Create a high-quality rendered image of your robot assembly for presentation or a design notebook.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape [Renderings Help](#) (<https://cad.onshape.com/help/Content/renderings.htm>).
 - **Task 2:** Collaborate with [Branches](#) and [Merge](#)
 - **Goal:** Create a branch to work on a new feature and then merge it back into the main workspace.
 - **Time Estimate:** 1 hour
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** Onshape [Branches and Merge Help](#) (<https://cad.onshape.com/help/Content/branches.htm>).
- **CAD.Course2.Session10 (Est. 3 hours):** Design Review and Documentation
 - **Task 1:** Robot Design Review
 - **Goal:** Present your complete CAD model to your team and judges for a design review, explaining your design choices and trade-offs.

- **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer with a web browser.
 - **Resource:** [FTC Game Manual 0: Design Strategy](#)
 - **Task 2:** Robot Design Documentation
 - **Goal:** Create a comprehensive design document that includes your CAD models, BOM, and a narrative of your design process.
 - **Time Estimate:** 1.5 hours
 - **Hardware Needed:** Computer.
 - **Resource:** [FTC Design Award Resources](#).
-

Combined CAD-CODE-COMPETE

Goal: To synthesize all skills by designing, building, and programming a complete robot for a past FTC game.

- **Combined.Session1 (Est. 3 days):** Robot Concept and CAD
 - **Task 1:** Game Analysis
 - **Goal:** Break down the rules and objectives of a past FTC game. Identify key scoring elements, field hazards, and robot requirements.
 - **Time Estimate:** 1 day
 - **Hardware Needed:** None.
 - **Resource:** [Analyzing and Breaking Down the 2026 FTC Game](#).
 - **Task 2:** Integrated Robot Concept
 - **Goal:** As a team, create a single document outlining the robot's design concept, including a sketch of the chassis and a description of the planned mechanisms.
 - **Time Estimate:** 1 day
 - **Hardware Needed:** None.
 - **Resource:** [Designing Our FTC Bot 2023-2024 - Instructables](#).
 - **Task 3:** Full Robot CAD
 - **Goal:** The CAD team must design the complete robot in Onshape, including the drivetrain and all subsystems, ensuring they fit together correctly and can be built with available parts.
 - **Time Estimate:** 1 day
 - **Hardware Needed:** Computer with a web browser.
- **Combined.Session2 (Est. 3 days):** Full Robot Code
 - **Task 1:** Full Robot Code
 - **Goal:** The software team must write a comprehensive program for the robot, including a robust tele-op and an autonomous mode that uses Road Runner and state machines to complete game objectives.
 - **Time Estimate:** 3 days
 - **Hardware Needed:** Computer, Driver Station phone, Gamepad, Control Hub.
- **Combined.Session3 (Est. 2 days):** Build and Test
 - **Task 1:** Final Build and Test
 - **Goal:** As a team, assemble the physical robot from the CAD models and test all its functions.
 - **Time Estimate:** 1.5 days
 - **Hardware Needed:** A complete set of physical robot parts, tools, computer, Control Hub, Driver Station phone, Gamepad.
 - **Task 2:** Final Competition Simulation

- **Goal:** Run the robot through a mock competition to evaluate its performance and prepare for a real event.
 - **Time Estimate:** 0.5 days
 - **Hardware Needed:** A complete robot, a full FTC field, computer, Control Hub, Driver Station phone, Gamepad.
-

Complete List of Resources

- **Software Resources:**
 - [Learn Java for FTC \(PDF\)](#)
 - [Road Runner Docs](#)
 - [Road Runner FAQs](#)
 - [GM0.org Mecanum Drive Tutorial](#)
 - [FTC SDK Samples on GitHub](#)
 - [FTC-Sim Courses](#)
 - [MeepMeep Trajectory Tool](#)
 - [FIRST Tech Challenge Programming Resources](#)
 - [FTC Driver Station Components](#)
- **CAD Resources:**
 - [Onshape Learning Center](#)
 - [Onshape Introduction to CAD Learning Path](#)
 - [Onshape Intro to Onshape Course](#)
 - [Onshape FTC Catalog](#)
 - [Onshape Article: Intro to Onshape](#)
 - [Onshape Article: Assemble a Robot](#)
 - [Onshape Article: CAD for Robotics Collaboration](#)
 - [Onshape Article: Communicating Your Design](#)
 - [Onshape Article: Creating Custom Components](#)
 - [Onshape Article: Manufacturing](#) I have updated the [Software.Course1](#) to have 10 sessions as you requested. I split the 3-hour session "Controlling Motors" into two new sessions: "Controlling a Single Motor" and "Motors and Hardware Mapping", and renumbered all subsequent sessions to maintain a logical progression.